CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2017 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **How will you create hyperlinks in a web document? Write the code also.**

HTML Hyperlink:

```
<a href="https://www.ccsuniversity.ac.in" target="_blank">Visit CCSU Official Portal</a>
```

Q2 (3M) **What is a Constructor? List its characteristics.**

Characteristics: Same name as class, no return type (not even void), automatically called on `new`, can be overloaded.

Q3 (3M) **Differentiate between Method Overloading and Method Overriding.**

Overloading: Same method name, different signatures, within same class (compile-time).
Overriding: Same signature, in subclass, overriding parent behavior (runtime).

Q4 (3M) **What is Bytecode in Java?**

Definition: Architecture-neutral intermediate instructions executed by JVM.

Q5 (3M) **What is the purpose of the 'this' keyword?**

Definition: Refers to the current class instance to differentiate shadowed instance fields from parameters.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Explain the Life Cycle of an Applet with diagram.**

Applet Lifecycle: Born (`init`) → Running (`start` , `paint`) → Idle (`stop`) → Dead (`destroy`).

Q7 (7.5M) **Explain Packages and Access Protection in Java.**

Access Protection: Detailed comparison of private, default, protected, and public scopes.

Q8 (7.5M) **Explain Interface in Java with code example.**

Interface: Blueprint of a class containing abstract methods and constants.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain Exception Handling in Java: try, catch, finally, throw, and throws with custom exception.**

Custom Exception:

```
class InvalidAgeException extends Exception {
    public InvalidAgeException(String msg) { super(msg); }
}
```

Q10 (15M) **Explain Multithreading in Java: Thread class, Runnable interface, and Thread Priorities.**

Multithreading: Full comparison of extending Thread vs implementing Runnable, lifecycle states, and priority scheduling.

Q11 (15M) **What is JDBC? Explain the 4 types of JDBC drivers and steps for database connectivity.**

JDBC Drivers: Type-1, Type-2, Type-3, and Type-4 drivers with architectural diagrams.

Q12 (15M) **Explain Servlets vs JSP. Write a complete HttpServlet program handling GET requests.**

Servlet Architecture: Complete HttpServlet implementation serving dynamic web responses.

Q13 (15M) **Explain Layout Managers in AWT: FlowLayout, BorderLayout, and GridLayout.**

Layout Managers: Visual layout structures and coordinate placements.
